Table 1: Per-benchmark results for the 100B-token MixtureVitae mixture composition ablation on the primary reasoning suite. Removing the instruction-and-reasoning subset causes a broad collapse in math, code, and reasoning that Phase 1 Tulu3 SFT/DPO does not recover, whereas removing the curated web subset leaves performance comparable to, and on several benchmarks stronger than, the full mixture at both 4K and 16K context. Protocol A (Appendix Table **??**). Values are mean$_{\pm SD}$ over decoding seeds 42, 1234, and 2025.

(a) Mathematics

Model	A24	A25	AMC	M500	GSM8K
MV-noinstruct	$0.0_{\pm 0.00}$	$0.0_{\pm 0.00}$	$0.0_{\pm 0.00}$	$0.4_{\pm 0.00}$	$2.1_{\pm 0.00}$
→ SFT	$0.0_{\pm 0.00}$	$0.0_{\pm 0.00}$	$0.8_{\pm 1.44}$	$2.3_{\pm 0.12}$	$3.9_{\pm 0.00}$
→ SFT → DPO	$0.0_{\pm 0.00}$	$0.0_{\pm 0.00}$	$4.2_{\pm 1.44}$	$2.6_{\pm 0.35}$	$4.0_{\pm 0.00}$
MV-noinstruct-16k	$0.0_{\pm 0.00}$	$0.0_{\pm 0.00}$	$0.0_{\pm 0.00}$	$1.9_{\pm 0.31}$	$4.2_{\pm 0.00}$
→ SFT	$0.0_{\pm 0.00}$	$0.0_{\pm 0.00}$	$0.0_{\pm 0.00}$	$1.8_{\pm 0.20}$	$4.5_{\pm 0.00}$
→ SFT → DPO	$0.0_{\pm 0.00}$	$0.0_{\pm 0.00}$	$0.0_{\pm 0.00}$	$2.4_{\pm 0.35}$	$4.8_{\pm 0.00}$
MV-noweb	$1.1_{\pm 1.92}$	$0.0_{\pm 0.00}$	$14.2_{\pm 3.82}$	$27.3_{\pm 0.64}$	$42.7_{\pm 0.00}$
→ SFT	$0.0_{\pm 0.00}$	$0.0_{\pm 0.00}$	$7.5_{\pm 0.00}$	$13.0_{\pm 0.20}$	$35.8_{\pm 0.00}$
→ SFT → DPO	$0.0_{\pm 0.00}$	$0.0_{\pm 0.00}$	$12.5_{\pm 0.00}$	$16.9_{\pm 0.46}$	$36.6_{\pm 0.00}$
MV-noweb-16k	$0.0_{\pm 0.00}$	$3.3_{\pm 0.00}$	$10.0_{\pm 0.00}$	$34.0_{\pm 0.69}$	$44.6_{\pm 0.00}$
→ SFT	$3.3_{\pm 0.00}$	$0.0_{\pm 0.00}$	$7.5_{\pm 0.00}$	$22.8_{\pm 0.53}$	$33.4_{\pm 0.00}$
→ SFT → DPO	$5.6_{\pm 1.92}$	$0.0_{\pm 0.00}$	$10.0_{\pm 0.00}$	$22.9_{\pm 1.17}$	$34.3_{\pm 0.00}$
MV	$0.0_{\pm 0.00}$	$0.0_{\pm 0.00}$	$2.5_{\pm 0.00}$	$14.3_{\pm 0.42}$	$42.7_{\pm 0.00}$
→ SFT	$0.0_{\pm 0.00}$	$0.0_{\pm 0.00}$	$5.8_{\pm 1.44}$	$10.3_{\pm 0.23}$	$37.8_{\pm 0.00}$
→ SFT → DPO	$0.0_{\pm 0.00}$	$0.0_{\pm 0.00}$	$6.7_{\pm 1.44}$	$12.0_{\pm 0.53}$	$38.7_{\pm 0.00}$
MV-16k	$0.0_{\pm 0.00}$	$0.0_{\pm 0.00}$	$9.2_{\pm 1.44}$	$33.9_{\pm 0.12}$	$40.3_{\pm 0.00}$
→ SFT	$0.0_{\pm 0.00}$	$0.0_{\pm 0.00}$	$0.0_{\pm 0.00}$	$5.3_{\pm 0.23}$	$36.6_{\pm 0.00}$
→ SFT → DPO	$0.0_{\pm 0.00}$	$0.0_{\pm 0.00}$	$0.0_{\pm 0.00}$	$5.2_{\pm 0.00}$	$37.1_{\pm 0.00}$

(b) Code, science and instruction following, and the average

Model	HE	LCB	MBPP	GPQA	JEE	IFE	Avg
MV-noinstruct	$0.0_{\pm 0.00}$	$0.0_{\pm 0.00}$	$14.5_{\pm 0.23}$	$23.1_{\pm 0.29}$	$0.0_{\pm 0.00}$	$30.8_{\pm 0.00}$	$6.4_{\pm 0.02}$
→ SFT	$12.8_{\pm 0.00}$	$0.5_{\pm 0.00}$	$14.9_{\pm 0.23}$	$23.1_{\pm 0.29}$	$2.5_{\pm 0.10}$	$35.0_{\pm 0.30}$	$8.7_{\pm 0.21}$
→ SFT → DPO	$14.0_{\pm 0.00}$	$0.1_{\pm 0.16}$	$14.3_{\pm 0.12}$	$22.4_{\pm 0.29}$	$3.9_{\pm 0.34}$	$36.7_{\pm 0.39}$	$9.3_{\pm 0.05}$
MV-noinstruct-16k	$10.4_{\pm 0.00}$	$0.0_{\pm 0.00}$	$12.3_{\pm 0.12}$	$25.4_{\pm 0.29}$	$5.0_{\pm 0.26}$	$37.0_{\pm 0.20}$	$8.8_{\pm 0.07}$
→ SFT	$15.0_{\pm 0.35}$	$0.2_{\pm 0.16}$	$15.9_{\pm 0.12}$	$25.9_{\pm 0.77}$	$2.1_{\pm 0.15}$	$40.4_{\pm 0.08}$	$9.6_{\pm 0.09}$
→ SFT → DPO	$14.4_{\pm 0.70}$	$0.0_{\pm 0.00}$	$15.5_{\pm 0.12}$	$26.3_{\pm 0.00}$	$2.0_{\pm 0.25}$	$41.9_{\pm 0.13}$	$9.8_{\pm 0.10}$
MV-noweb	$43.7_{\pm 0.35}$	$4.8_{\pm 0.41}$	$33.0_{\pm 0.00}$	$19.9_{\pm 0.77}$	$3.1_{\pm 0.25}$	$31.3_{\pm 0.26}$	$20.1_{\pm 0.43}$
→ SFT	$44.1_{\pm 0.35}$	$0.9_{\pm 0.16}$	$37.3_{\pm 0.31}$	$26.3_{\pm 0.00}$	$3.2_{\pm 0.34}$	$40.5_{\pm 0.20}$	$19.0_{\pm 0.04}$
→ SFT → DPO	$44.1_{\pm 0.35}$	$0.6_{\pm 0.16}$	$37.1_{\pm 0.23}$	$25.6_{\pm 0.29}$	$4.3_{\pm 0.28}$	$40.8_{\pm 0.23}$	$19.9_{\pm 0.11}$
MV-noweb-16k	$39.0_{\pm 0.00}$	$2.1_{\pm 0.41}$	$36.4_{\pm 0.00}$	$21.7_{\pm 0.00}$	$6.0_{\pm 0.46}$	$41.4_{\pm 0.27}$	$21.7_{\pm 0.09}$
→ SFT	$41.3_{\pm 0.35}$	$1.9_{\pm 0.27}$	$34.7_{\pm 0.12}$	$23.1_{\pm 0.29}$	$3.6_{\pm 0.32}$	$43.5_{\pm 0.49}$	$19.6_{\pm 0.09}$
→ SFT → DPO	$39.8_{\pm 0.35}$	$1.6_{\pm 0.27}$	$35.7_{\pm 0.12}$	$23.2_{\pm 0.00}$	$3.7_{\pm 0.07}$	$43.8_{\pm 0.91}$	$20.1_{\pm 0.18}$
MV	$0.0_{\pm 0.00}$	$0.0_{\pm 0.00}$	$0.0_{\pm 0.00}$	$22.4_{\pm 2.59}$	$3.0_{\pm 0.39}$	$26.7_{\pm 0.00}$	$10.1_{\pm 0.28}$
→ SFT	$41.9_{\pm 0.35}$	$2.7_{\pm 0.00}$	$35.6_{\pm 0.20}$	$17.2_{\pm 0.51}$	$2.8_{\pm 0.49}$	$40.9_{\pm 0.40}$	$17.7_{\pm 0.09}$
→ SFT → DPO	$41.5_{\pm 0.00}$	$2.5_{\pm 0.16}$	$36.2_{\pm 0.20}$	$19.5_{\pm 0.29}$	$1.8_{\pm 0.17}$	$43.3_{\pm 0.15}$	$18.4_{\pm 0.15}$
MV-16k	$37.8_{\pm 0.00}$	$3.8_{\pm 0.27}$	$35.1_{\pm 0.31}$	$24.9_{\pm 0.29}$	$8.6_{\pm 0.18}$	$44.0_{\pm 0.40}$	$21.6_{\pm 0.10}$
→ SFT	$38.4_{\pm 0.00}$	$0.5_{\pm 0.16}$	$34.8_{\pm 0.00}$	$24.2_{\pm 0.00}$	$4.1_{\pm 0.06}$	$43.8_{\pm 0.15}$	$17.1_{\pm 0.02}$
→ SFT → DPO	$40.9_{\pm 0.00}$	$0.4_{\pm 0.16}$	$35.5_{\pm 0.42}$	$23.7_{\pm 0.00}$	$3.7_{\pm 0.34}$	$45.0_{\pm 0.20}$	$17.4_{\pm 0.05}$

Table 2: Reasoning suite results for the smaller-scale comparison between MV (0.4B) and Baguettotron. Across 4K and 16K context, MV is substantially more receptive to Phase 1 post-training, especially on code and broader reasoning, while Baguettotron remains weak throughout. Protocol A (Appendix Table **??**). Values are mean$_{\pm SD}$ over decoding seeds 42, 1234, and 2025.

(a) Mathematics

Model	A24	A25	AMC	M500	GSM8K
Baguettotron	$0.0_{\pm 0.00}$	$0.0_{\pm 0.00}$	$0.0_{\pm 0.00}$	$0.5_{\pm 0.31}$	$2.4_{\pm 0.00}$
→ SFT	$0.0_{\pm 0.00}$	$0.0_{\pm 0.00}$	$0.0_{\pm 0.00}$	$2.1_{\pm 0.23}$	$8.0_{\pm 0.00}$
→ SFT → DPO	$0.0_{\pm 0.00}$	$0.0_{\pm 0.00}$	$3.3_{\pm 1.44}$	$2.8_{\pm 0.20}$	$7.7_{\pm 0.00}$
Baguettotron 16k	$0.0_{\pm 0.00}$	$0.0_{\pm 0.00}$	$0.0_{\pm 0.00}$	$1.8_{\pm 0.20}$	$5.4_{\pm 0.00}$
→ SFT	$0.0_{\pm 0.00}$	$0.0_{\pm 0.00}$	$5.0_{\pm 0.00}$	$2.0_{\pm 0.20}$	$8.3_{\pm 0.00}$
→ SFT → DPO	$0.0_{\pm 0.00}$	$0.0_{\pm 0.00}$	$2.5_{\pm 0.00}$	$2.9_{\pm 0.23}$	$8.3_{\pm 0.00}$
MV (0.4B)	$0.0_{\pm 0.00}$	$0.0_{\pm 0.00}$	$2.5_{\pm 0.00}$	$1.0_{\pm 0.00}$	$29.2_{\pm 0.00}$
→ SFT	$0.0_{\pm 0.00}$	$0.0_{\pm 0.00}$	$0.0_{\pm 0.00}$	$9.1_{\pm 0.12}$	$23.4_{\pm 0.00}$
→ SFT → DPO	$0.0_{\pm 0.00}$	$0.0_{\pm 0.00}$	$2.5_{\pm 0.00}$	$11.7_{\pm 0.23}$	$23.4_{\pm 0.00}$
MV-16k (0.4B)	$0.0_{\pm 0.00}$	$0.0_{\pm 0.00}$	$8.3_{\pm 1.44}$	$19.6_{\pm 0.00}$	$29.4_{\pm 0.00}$
→ SFT	$0.0_{\pm 0.00}$	$0.0_{\pm 0.00}$	$2.5_{\pm 0.00}$	$9.4_{\pm 0.20}$	$23.6_{\pm 0.00}$
→ SFT → DPO	$0.0_{\pm 0.00}$	$0.0_{\pm 0.00}$	$5.0_{\pm 0.00}$	$10.1_{\pm 0.42}$	$25.0_{\pm 0.00}$

(b) Code, science and instruction following, and the average

Model	HE	LCB	MBPP	GPQA	JEE	IFE	Avg
Baguettotron	$0.0_{\pm 0.00}$	$0.0_{\pm 0.00}$	$0.0_{\pm 0.00}$	$21.9_{\pm 2.10}$	$0.2_{\pm 0.00}$	$33.7_{\pm 0.60}$	$5.3_{\pm 0.12}$
→ SFT	$0.0_{\pm 0.00}$	$0.0_{\pm 0.00}$	$0.0_{\pm 0.00}$	$26.3_{\pm 0.00}$	$1.7_{\pm 0.06}$	$36.2_{\pm 0.27}$	$6.8_{\pm 0.02}$
→ SFT → DPO	$0.0_{\pm 0.00}$	$0.0_{\pm 0.00}$	$0.0_{\pm 0.00}$	$25.3_{\pm 0.00}$	$1.6_{\pm 0.15}$	$36.8_{\pm 0.30}$	$7.0_{\pm 0.12}$
Baguettotron 16k	$1.2_{\pm 0.00}$	$0.0_{\pm 0.00}$	$0.0_{\pm 0.00}$	$21.0_{\pm 0.29}$	$1.8_{\pm 0.22}$	$40.4_{\pm 0.53}$	$6.5_{\pm 0.06}$
→ SFT	$0.0_{\pm 0.00}$	$0.0_{\pm 0.00}$	$0.2_{\pm 0.00}$	$30.3_{\pm 0.00}$	$1.4_{\pm 0.41}$	$37.3_{\pm 0.33}$	$7.7_{\pm 0.06}$
→ SFT → DPO	$0.0_{\pm 0.00}$	$0.0_{\pm 0.00}$	$0.0_{\pm 0.00}$	$29.8_{\pm 0.00}$	$1.9_{\pm 0.25}$	$37.5_{\pm 0.40}$	$7.5_{\pm 0.04}$
MV (0.4B)	$25.6_{\pm 0.00}$	$0.5_{\pm 0.00}$	$8.6_{\pm 0.00}$	$22.9_{\pm 0.58}$	$0.8_{\pm 0.13}$	$25.6_{\pm 0.00}$	$10.6_{\pm 0.04}$
→ SFT	$28.0_{\pm 0.00}$	$2.7_{\pm 0.00}$	$22.4_{\pm 0.35}$	$27.3_{\pm 0.00}$	$2.3_{\pm 0.08}$	$32.5_{\pm 0.00}$	$13.4_{\pm 0.04}$
→ SFT → DPO	$29.3_{\pm 0.00}$	$2.7_{\pm 0.00}$	$21.6_{\pm 0.20}$	$27.8_{\pm 0.00}$	$2.0_{\pm 0.19}$	$34.3_{\pm 0.00}$	$14.1_{\pm 0.03}$
MV-16k (0.4B)	$23.2_{\pm 0.00}$	$1.1_{\pm 0.00}$	$22.5_{\pm 0.23}$	$25.8_{\pm 0.00}$	$4.1_{\pm 0.82}$	$35.2_{\pm 0.00}$	$15.4_{\pm 0.07}$
→ SFT	$28.7_{\pm 0.00}$	$1.2_{\pm 0.16}$	$20.5_{\pm 0.12}$	$24.1_{\pm 0.29}$	$2.8_{\pm 0.27}$	$37.7_{\pm 0.00}$	$13.7_{\pm 0.06}$
→ SFT → DPO	$28.0_{\pm 0.00}$	$1.4_{\pm 0.00}$	$20.6_{\pm 0.00}$	$22.6_{\pm 0.29}$	$2.1_{\pm 0.10}$	$38.8_{\pm 0.00}$	$14.0_{\pm 0.02}$

Table 3: Reasoning suite of 300B MixtureVitae checkpoints trained without and with the historical 13-gram filtering of their data, at 4K and 16K context, before and after Phase 1 Tulu3 SFT/DPO, a check of that procedure, not of benchmark-disjoint data (Appendix **??**). Protocol A (Appendix Table **??**). Values are mean$_{\pm SD}$ over decoding seeds 42, 1234, and 2025.

(a) Mathematics

	Model	A24	A25	AMC	M500	GSM8K
Without filtering	MV	$0.0_{\pm 0.00}$	$0.0_{\pm 0.00}$	$13.3_{\pm 1.44}$	$13.7_{\pm 0.23}$	$53.6_{\pm 0.00}$
	→ SFT	$3.3_{\pm 0.00}$	$0.0_{\pm 0.00}$	$10.0_{\pm 2.50}$	$19.7_{\pm 0.81}$	$50.0_{\pm 0.00}$
	→ SFT → DPO	$1.1_{\pm 1.92}$	$0.0_{\pm 0.00}$	$10.0_{\pm 0.00}$	$20.3_{\pm 0.42}$	$51.1_{\pm 0.00}$
	MV-16k	$1.1_{\pm 1.92}$	$0.0_{\pm 0.00}$	$16.7_{\pm 2.89}$	$38.6_{\pm 0.35}$	$62.0_{\pm 0.00}$
	→ SFT	$0.0_{\pm 0.00}$	$0.0_{\pm 0.00}$	$0.0_{\pm 0.00}$	$25.3_{\pm 0.12}$	$51.8_{\pm 0.00}$
	→ SFT → DPO	$0.0_{\pm 0.00}$	$0.0_{\pm 0.00}$	$0.0_{\pm 0.00}$	$27.3_{\pm 0.42}$	$52.2_{\pm 0.00}$
With filtering	MV	$0.0_{\pm 0.00}$	$7.8_{\pm 5.09}$	$5.0_{\pm 0.00}$	$17.8_{\pm 0.72}$	$54.1_{\pm 0.00}$
	→ SFT	$0.0_{\pm 0.00}$	$0.0_{\pm 0.00}$	$12.5_{\pm 0.00}$	$29.5_{\pm 0.23}$	$49.9_{\pm 0.00}$
	→ SFT → DPO	$0.0_{\pm 0.00}$	$3.3_{\pm 0.00}$	$10.8_{\pm 1.44}$	$30.5_{\pm 0.12}$	$50.2_{\pm 0.00}$
	MV-16k	$1.1_{\pm 1.92}$	$3.3_{\pm 0.00}$	$23.3_{\pm 8.04}$	$47.0_{\pm 0.53}$	$60.0_{\pm 0.00}$
	→ SFT	$3.3_{\pm 0.00}$	$3.3_{\pm 0.00}$	$15.0_{\pm 0.00}$	$35.0_{\pm 0.53}$	$51.6_{\pm 0.00}$
	→ SFT → DPO	$3.3_{\pm 0.00}$	$1.1_{\pm 1.92}$	$17.5_{\pm 0.00}$	$37.0_{\pm 0.80}$	$52.6_{\pm 0.00}$

(b) Code, science and instruction following, and the average

	Model	HE	LCB	MBPP	GPQA	JEE	IFE	Avg
Without filtering	MV	$46.5_{\pm 0.35}$	$6.3_{\pm 0.41}$	$43.5_{\pm 0.42}$	$25.1_{\pm 1.27}$	$5.6_{\pm 0.29}$	$36.4_{\pm 0.08}$	$22.2_{\pm 0.14}$
	→ SFT	$51.4_{\pm 0.35}$	$7.0_{\pm 0.31}$	$42.7_{\pm 0.12}$	$26.8_{\pm 0.87}$	$3.0_{\pm 0.40}$	$49.3_{\pm 0.27}$	$23.9_{\pm 0.21}$
	→ SFT → DPO	$56.3_{\pm 0.70}$	$7.5_{\pm 0.16}$	$43.7_{\pm 0.12}$	$27.8_{\pm 0.00}$	$3.3_{\pm 0.23}$	$51.8_{\pm 0.64}$	$24.8_{\pm 0.22}$
	MV-16k	$51.0_{\pm 0.35}$	$7.7_{\pm 0.56}$	$42.0_{\pm 0.35}$	$27.9_{\pm 1.27}$	$13.4_{\pm 0.59}$	$47.7_{\pm 0.38}$	$28.0_{\pm 0.29}$
	→ SFT	$52.2_{\pm 0.35}$	$5.7_{\pm 0.00}$	$43.5_{\pm 0.12}$	$24.6_{\pm 0.29}$	$4.1_{\pm 0.47}$	$56.0_{\pm 0.38}$	$23.9_{\pm 0.08}$
	→ SFT → DPO	$52.8_{\pm 0.35}$	$6.6_{\pm 0.16}$	$42.5_{\pm 0.42}$	$23.4_{\pm 0.77}$	$3.4_{\pm 0.20}$	$56.5_{\pm 0.78}$	$24.1_{\pm 0.03}$
With filtering	MV	$13.6_{\pm 0.93}$	$7.7_{\pm 0.16}$	$19.5_{\pm 0.12}$	$24.4_{\pm 0.77}$	$2.9_{\pm 0.51}$	$34.7_{\pm 0.13}$	$17.0_{\pm 0.38}$
	→ SFT	$44.3_{\pm 0.70}$	$1.2_{\pm 0.16}$	$39.9_{\pm 0.23}$	$20.7_{\pm 0.87}$	$3.7_{\pm 0.55}$	$48.0_{\pm 0.08}$	$22.7_{\pm 0.14}$
	→ SFT → DPO	$47.4_{\pm 0.35}$	$1.0_{\pm 0.16}$	$41.2_{\pm 0.20}$	$21.2_{\pm 0.00}$	$4.5_{\pm 0.34}$	$49.8_{\pm 0.08}$	$23.6_{\pm 0.17}$
	MV-16k	$54.1_{\pm 0.35}$	$11.5_{\pm 0.16}$	$40.1_{\pm 0.23}$	$23.4_{\pm 0.29}$	$8.0_{\pm 0.24}$	$47.7_{\pm 0.34}$	$29.1_{\pm 0.69}$
	→ SFT	$54.1_{\pm 0.35}$	$9.7_{\pm 0.16}$	$40.4_{\pm 0.35}$	$23.4_{\pm 0.29}$	$5.1_{\pm 0.27}$	$51.9_{\pm 0.46}$	$26.6_{\pm 0.11}$
	→ SFT → DPO	$52.6_{\pm 0.70}$	$9.4_{\pm 0.31}$	$40.3_{\pm 0.12}$	$23.4_{\pm 0.29}$	$6.2_{\pm 0.20}$	$52.6_{\pm 0.60}$	$26.9_{\pm 0.04}$

Table 4: Reasoning suite with vs. without prompt-token loss masking in Phase-1 post-training (1.7B MixtureVitae 300BT, decontaminated). The rows with masking are the `laion` masked checkpoints of Table **??**, and the rows without masking are the main MV checkpoints of Table 3. Masking raises the suite average by 0.4 to 1.4 points at every stage, and these differences do not affect our comparison. Protocol A (Appendix Table **??**). Values are mean$_{\pm \mathrm{SD}}$ over decoding seeds 42, 1234, and 2025.

(a) Mathematics

	Model	A24	A25	AMC	M500	GSM8K
With masking	MV					
	→ SFT	$0.0_{\pm 0.00}$	$0.0_{\pm 0.00}$	$10.8_{\pm 3.82}$	$27.0_{\pm 0.35}$	$52.2_{\pm 0.00}$
	→ SFT → DPO	$0.0_{\pm 0.00}$	$0.0_{\pm 0.00}$	$11.7_{\pm 1.44}$	$28.1_{\pm 0.42}$	$52.7_{\pm 0.00}$
	MV-16k					
	→ SFT	$0.0_{\pm 0.00}$	$0.0_{\pm 0.00}$	$10.0_{\pm 0.00}$	$35.5_{\pm 0.23}$	$57.5_{\pm 0.00}$
	→ SFT → DPO	$0.0_{\pm 0.00}$	$0.0_{\pm 0.00}$	$15.0_{\pm 0.00}$	$35.7_{\pm 0.46}$	$57.8_{\pm 0.00}$
Without masking	MV					
	→ SFT	$0.0_{\pm 0.00}$	$0.0_{\pm 0.00}$	$12.5_{\pm 0.00}$	$29.5_{\pm 0.23}$	$49.9_{\pm 0.00}$
	→ SFT → DPO	$0.0_{\pm 0.00}$	$3.3_{\pm 0.00}$	$10.8_{\pm 1.44}$	$30.5_{\pm 0.12}$	$50.2_{\pm 0.00}$
	MV-16k					
	→ SFT	$3.3_{\pm 0.00}$	$3.3_{\pm 0.00}$	$15.0_{\pm 0.00}$	$35.0_{\pm 0.53}$	$51.6_{\pm 0.00}$
	→ SFT → DPO	$3.3_{\pm 0.00}$	$1.1_{\pm 1.92}$	$17.5_{\pm 0.00}$	$37.0_{\pm 0.80}$	$52.6_{\pm 0.00}$

(b) Code, science and instruction following, and the average

	Model	HE	LCB	MBPP	GPQA	JEE	IFE	Avg
With masking	MV							
	→ SFT	$53.7_{\pm 0.00}$	$2.3_{\pm 0.41}$	$42.6_{\pm 0.00}$	$19.5_{\pm 0.58}$	$5.0_{\pm 0.46}$	$51.7_{\pm 0.08}$	$24.1_{\pm 0.36}$
	→ SFT → DPO	$53.0_{\pm 0.61}$	$2.8_{\pm 0.41}$	$42.1_{\pm 0.31}$	$20.7_{\pm 0.51}$	$4.0_{\pm 0.33}$	$54.6_{\pm 0.00}$	$24.5_{\pm 0.20}$
	MV-16k							
	→ SFT	$56.7_{\pm 0.00}$	$8.1_{\pm 0.27}$	$41.0_{\pm 0.40}$	$25.8_{\pm 0.00}$	$7.2_{\pm 0.74}$	$55.5_{\pm 0.26}$	$27.0_{\pm 0.09}$
	→ SFT → DPO	$54.7_{\pm 0.70}$	$8.1_{\pm 0.27}$	$41.3_{\pm 0.31}$	$26.9_{\pm 0.29}$	$6.8_{\pm 0.34}$	$56.4_{\pm 1.03}$	$27.5_{\pm 0.15}$
Without masking	MV							
	→ SFT	$44.3_{\pm 0.70}$	$1.2_{\pm 0.16}$	$39.9_{\pm 0.23}$	$20.7_{\pm 0.87}$	$3.7_{\pm 0.55}$	$48.0_{\pm 0.08}$	$22.7_{\pm 0.14}$
	→ SFT → DPO	$47.4_{\pm 0.35}$	$1.0_{\pm 0.16}$	$41.2_{\pm 0.20}$	$21.2_{\pm 0.00}$	$4.5_{\pm 0.34}$	$49.8_{\pm 0.08}$	$23.6_{\pm 0.17}$
	MV-16k							
	→ SFT	$54.1_{\pm 0.35}$	$9.7_{\pm 0.16}$	$40.4_{\pm 0.35}$	$23.4_{\pm 0.29}$	$5.1_{\pm 0.27}$	$51.9_{\pm 0.46}$	$26.6_{\pm 0.11}$
	→ SFT → DPO	$52.6_{\pm 0.70}$	$9.4_{\pm 0.31}$	$40.3_{\pm 0.12}$	$23.4_{\pm 0.29}$	$6.2_{\pm 0.20}$	$52.6_{\pm 0.60}$	$26.9_{\pm 0.04}$